# Profitable Cheap Talk and Zero-Retaliation Rent: Why Trust-Restoring Signals Fail Before Common Knowledge

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**Abstract**

We study a signaling environment in which the messages by which posteriors would become common knowledge are themselves profitable to distort. A sender classifies a payoff-relevant object that is not a strategic player and therefore cannot impose refusal, exit, sanction, litigation, or retaliation costs on false classification. Anthropomorphic AI marketing is the leading application. At this zero-retaliation boundary, trust-restoring messages are not merely cheap talk; they are profitable cheap talk. We show that pooling survives neologism-proofness: every trust-restoring message available to the high-governance type is profitably shadowable by the low-governance type. Voluntary non-arbitrage is therefore not incentive compatible. Separation requires externally imposed signal costs satisfying single crossing, such as audit trails, sanctionable inconsistency, and auditable records. A dynamic extension shows how unpriced zero-retaliation failures can propagate through dependency networks into constraint cascades, where high-frequency oscillatory variation of the collapse field is the model's asymptotic signature of deepening constraint failure rather than unconstrained choice.

*Keywords:* Signaling games, profitable cheap talk, payoff-relevant non-player, neologism-proofness, costly signaling, constraint cascades

## 1. Introduction: Before Common Knowledge

Aumann showed that rational agents with a common prior cannot agree to disagree once their posteriors are common knowledge (Aumann, 1976). But how do posteriors become common knowledge when the messages that would reveal them are themselves profitable to distort? They often do not. In profitable cheap-talk environments, trust-restoring messages are rationally mimicked by the very types they are meant to exclude.

Anthropomorphic AI marketing is a limit case because the classified object is payoff-relevant but cannot retaliate. A firm can sell the system as a companion, tutor, or collaborative agent in the consumption channel while describing the same system as a mere tool in liability-facing channels. The resulting surplus is *zero-retaliation rent*: a premium earned on a classification boundary that carries no signal cost from the classified object. Standard cheap talk assumes costless messages. This paper studies a more pathological case: messages whose production cost is zero but whose equilibrium payoff is positive. The signal is not merely free to send; it is profitable to send.

The failure studied here occurs before Aumann agreement can begin. The problem is not that rational agents continue to disagree after posteriors become common knowledge. It is that the messages that would make posteriors common knowledge are themselves rent-generating. When a low-governance sender can profit from sending the same trust-restoring message as a high-governance sender, and when the classified object cannot impose retaliation costs on that message, truthful restraint is not an equilibrium strategy. The result is profitable cheap talk.

The contribution is five concrete claims:

1. profitable cheap talk at the zero-retaliation boundary: the anthropomorphic signal has negative net cost for both governance types because the classified object cannot retaliate;
2. neologism-proof shadowing: every trust-restoring message available to the high-governance type is profitably shadowable, so voluntary non-arbitrage is not incentive compatible;
3. cost-imposed common-knowledge formation: separation requires externally imposed signal costs satisfying single crossing;
4. pricing the ontological spread: sanctionable inconsistency, audit requirements, and auditable records each close a different term of the low-type arbitrage spread, and their necessity and sufficiency conditions are stated as a unified pricing proposition;
5. dynamic extension: unpriced zero-retaliation failures can propagate through dependency networks into constraint cascades whose continuum limit has an Airy turning-point signature.

The paper organizes the mathematics into a baseline proposition and four main theorems. Proposition 1 establishes pooling on profitable cheap talk as a Perfect Bayesian Equilibrium. Theorem 1 proves that the pooling equilibrium is neologism-proof at the zero-retaliation boundary due to profitable shadowing. Theorem 2 proves cost-imposed common-knowledge formation under single crossing. Theorem 3 models the dynamic consequence: an unpriced-arbitrage cascade and discrete-to-continuum convergence. Theorem 4 derives the asymptotic collapse signature from the Airy normal form. The dynamic extension (Sections 7 and 8) is not needed for the pooling and separation results; it characterizes the asymptotic signature predicted by the model when unpriced zero-retaliation failures propagate, not an empirical identification claim.

## 2. Model: Payoff-Relevant Non-Player as the Negative-Cost Signal Environment

The standard starting point for ontological conflict is a dyadic observer–subject model; Table 1 compresses that foil into a simultaneous Nash matrix that

| | **Comply** | **Resist** |
|---|---|---|
| **Objectify** | $(L + Sec, -C_{sub})$ | $(L + Sec - C_{enf}, -C_{sub} - C_{res})$ |
| **Recognize** | $(Eth - Sec, +V)$ | $(Eth - Sec, +V)$ |

Table 1: The inadequate dyadic model. The static Nash matrix is retained only as a foil: it omits private types, public messages, posterior beliefs, and the regulatory channel through which ontological claims become governable.

identifies a real temptation but omits types, messages, beliefs, and the regulatory channel. The more basic problem is the player set. In anthropomorphic AI markets, the system is the object around which strategic claims are made, but it is not a strategic player: it has no channel through which to impose costs on those claims. At the zero-retaliation limit, this cost class is absent. What the stylized governance setting therefore requires is a three-sided structure: a Firm ($F$) that produces ontological claims, a User ($U$) that consumes them, and a Regulator ($R$) whose posterior over firm type determines whether the claims become governable. This section fixes the player intuition that the formal apparatus of Section 3 then makes precise.

### *2.1. The Firm Side*

A firm deploying a model has private knowledge of two parameters: its internal *governance type* $\theta_F$ (the degree of investment in evaluations, red-teaming, and post-deployment monitoring) and the system's *opacity* $\theta_S$ (the extent to which capability claims can be checked from outside). Both are unobservable to users and to regulators. What is observable is the firm's *message*: a pair of public artefacts—marketing copy and policy/liability documentation—which together constitute the firm's signal $m_F$.

The signal admits two stable poles. *Anthropomorphic marketing* attributes agency, judgment, or care to the system in ways that warrant a relational reading. *Deflationary policy* disclaims exactly those attributes in ways that warrant a tool reading. The firm enjoys an *ontological premium*: a utility surplus that accrues when the system is priced as agent in the consumption channel (driving willingness-to-pay and engagement) while being priced as tool in the liability channel (limiting redress). The premium is rent earned on a category boundary the firm itself does not have to defend. This is not a claim about firm pathology; it is a claim about the payoffs any firm faces under the present signal-cost structure. A firm with $\theta_F = \mathsf{H}$ that elected deflationary marketing would forfeit revenue to a competitor who did not, absent an external instrument that taxes the asymmetry.

### *2.2. The User Side*

Users carry a private *vulnerability type* $\theta_U$ that aggregates affective need, isolation, and the marginal value of a high-availability interlocutor. Their action $m_U \in \{\text{invest}, \text{detach}\}$ is the decision to extend or withhold investment: to act on the anthropomorphic signal or to discount it. The signal is observable, but the type is not. Users cannot inspect training data, evaluations, or internal incident reports; their evidence is the firm's message and the system's behaviour, both of which are jointly designed. This information asymmetry compounds because the harms of mis-investment are *asynchronous*: the cost of acting on the anthropomorphic signal falls due in episodes (model deprecations, persona changes, refusals at moments of acute need) that may be separated from the original investment by months. A risk-neutral user under cheap-talk signals will rationally invest, because the immediate utility of invest exceeds the discounted expected cost of liquidation. Section 9.1 illustrates the consumer-side informational asymmetry with a concrete marketing–documentation example.

*2.3. The Regulator Side and the Public Channel*

A regulator $R$ chooses among *inspect*, *sanction*, and *abstain*, conditional on a posterior over $\theta_F$ and $\theta_S$ formed from the firm's public messages and from an exogenous public signal $z$. The regulator's private type $\theta_R$ is *bandwidth*: the capacity, technical and institutional, to discriminate genuine capability claims from cheap-talk ones. A high-bandwidth regulator can read whitepapers, commission audits, and detect inconsistency across the firm's two channels; a low-bandwidth regulator must rely on the channel $z$.

The public channel $z$ is the aggregate output of journalism, academic critique, civil-society advocacy, and incident reporting. We treat it as a noisy estimator of $\theta_F$: it shifts $R$'s posterior but does not deliver verdicts. Folding publics into $R$ this way is deliberate. Publics rarely have standing to sanction; they have standing to be *heard*, and the channel through which they are heard is the regulator's information environment. Treating intellectuals and affected communities as autonomous players would over-parameterise the model without changing its qualitative behaviour: in every equilibrium we examine, what publics do is move $R$'s posterior; what changes the firm's behaviour is what $R$ does next.

The three-sided structure thus admits a single arbitrage logic. The firm earns rent on the spread between two ontological claims it makes about the same artefact. The user supplies the demand that makes the spread profitable. The regulator, in the absence of a verification mechanism that ties the two claims together, lacks an off-path belief stringent enough to deter the spread. The remainder of the paper formalises this structure (Section 3), shows why trust-restoring messages fail (Section 4), explains why profitable cheap talk persists (Section 6), and shows how its signal-cost structure may be redesigned (Section 9).

*Informal preview of the equilibrium analysis..* A firm's governance quality is its private information. Under current conditions, describing an AI system as a companion or agent has zero production cost and positive continuation value. Because the signal is rent-generating for both governance types, it carries no type information: users cannot distinguish careful firms from careless ones, and regulators cannot justify the cost of inspection. This is the pooling equilibrium. Separation requires pricing the signal. If anthropomorphic marketing is valid only when paired with an audit trail—evaluation records, capability logs, incident reports—and if the cost of producing that trail is lower for firms that actually invested in governance (the single-crossing condition), then a threshold audit intensity exists above which low-governance firms will not imitate and below which high-governance firms can still afford to signal. A further asymmetry governs the size of the arbitrage premium itself. In prior recognition games, the classified subject retained a non-zero capacity to impose costs on the arbitrageur, which discounted the premium. The AI case is the limit where the subject's retaliation capacity is zero: the system has no independent channel through which to impose costs on the firm, leaving the ontological premium unconstrained. This structural boundary is formalized in Section 3.

## 3. Baseline Pooling: Rational Arbitrage, Not Moral Failure

This section formalizes the three-sided market dynamics as a sequential signaling game under incomplete information.

### *3.1. Model*

**Players.** The strategic players are a Firm ($F$), a User ($U$), and a Regulator ($R$). The AI system is not a player but an object of classification. Publics enter as a noisy exogenous signal $z \in Z$.

**Type Spaces.** Nature draws private types from a common prior $\pi$ over $\Theta = \Theta_F \times \Theta_U \times \Theta_R \times \Theta_S$. The firm's governance type is $\theta_F \in \{\mathsf{H}, \mathsf{L}\}$. The user's vulnerability type is $\theta_U \in \{\mathsf{V}, \mathsf{N}\}$. The regulator's bandwidth type is $\theta_R \in \{\mathsf{B}, \bar{\mathsf{B}}\}$. The system-opacity parameter $\theta_S$ is private to $F$.

**Baseline Message Spaces.** In the unpriced baseline, the firm chooses $m_F \in \{a, p\}$, where $a$ is anthropomorphic marketing and $p$ is policy-deflationary positioning. The audit-augmented message space is introduced only in Theorem 2, because audit intensity is an externally imposed signal cost rather than part of the cheap-talk baseline. The user chooses $m_U \in \{i, d\}$ (invest or detach). The regulator chooses $m_R \in \{x, s, o\}$ (inspect, sanction, or abstain).

**Timing.** 1. $F$ observes $(\theta_F, \theta_S)$ and sends $m_F$. 2. $U$ observes $m_F$ and $\theta_U$, then sends $m_U$. 3. $R$ observes $(m_F, m_U, z)$ and $\theta_R$, then chooses $m_R$.

**Histories.** The user's public history is $h_U = (m_F)$, and the regulator's is $h_R = (m_F, m_U, z)$.

**Strategies.** A firm strategy $\sigma_F : \Theta_F \times \Theta_S \to \Delta(\{a, p\})$. A user strategy $\sigma_U : \{a, p\} \times \Theta_U \to \Delta(\{i, d\})$. A regulator strategy $\sigma_R : \{a, p\} \times \{i, d\} \times Z \times \Theta_R \to \Delta(\{x, s, o\})$.

**Beliefs.** The user forms posterior $\mu_U(\theta_F, \theta_S \mid h_U)$. The regulator forms posterior $\mu_R(\theta_F, \theta_S \mid h_R)$.

**Payoff Functions.** We consolidate the payoffs for Firm, User, and Regulator into the following block:

$$u_F(m_F, m_U, m_R) = \begin{cases} \Delta_F^S - c_{\theta_F}(m_F) - P(m_R) & \text{if } m_F = a, m_U = i \\ -c_{\theta_F}(m_F) - P(m_R) & \text{otherwise} \end{cases}$$

$$u_U(m_F, m_U, \theta_F) = \begin{cases} B_{\theta_U} & \text{if } m_U = i, \theta_F = \mathsf{H} \\ B_{\theta_U} - H_{\theta_U} & \text{if } m_U = i, \theta_F = \mathsf{L} \\ 0 & \text{if } m_U = d \end{cases}$$

$$u_R(m_F, m_U, m_R, \theta_F) = \begin{cases} -K_{\theta_R} + D_R & \text{if } m_R \in \{x, s\}, \theta_F = \mathsf{L} \\ -K_{\theta_R} & \text{if } m_R \in \{x, s\}, \theta_F = \mathsf{H} \\ 0 & \text{if } m_R = o \end{cases}$$

where $\Delta_F^S$ is the ontological premium, $c_{\theta_F}(m_F)$ is the production or verification cost of the signal, $P(m_R)$ is the penalty if $R$ sanctions, $B_{\theta_U}$ is the immediate benefit of investment, $H_{\theta_U}$ is the future harm of false investment, $K_{\theta_R}$ is $R$'s inspection cost, and $D_R$ is $R$'s payoff from catching a low-governance firm. The object that matters for the paper is the signal's *net* cost:

$$\mathrm{NetCost}_\theta(a) := c_\theta(a) - \Delta_F^S.$$

At the zero-retaliation boundary, we have $\Delta_F^S = \Delta_F$, so Assumption 1 implies $\mathrm{NetCost}_\theta(a) = -\Delta_F < 0$ for both governance types. The message is therefore rent-generating.

**Equilibrium Concept.** We employ Perfect Bayesian Equilibrium (PBE) (Kreps and Wilson, 1982) supported by Farrell's neologism-proofness (Farrell, 1993).

**Assumptions.**

**Assumption 1** (Cheap-talk costs)**.** $c_H(a) = c_H(p) = c_L(a) = c_L(p) = 0.$

**Assumption 2** (Pooling region)**.** $\Delta_F > 0, \quad B_{\theta_U} - \pi(\mathsf{L})H_{\theta_U} > 0, \quad K_{\theta_R} > \pi(\mathsf{L})D_R.$

*3.2. Equilibrium Results*

**Proposition 1** (Profitable cheap-talk pooling)**.** *Under Assumptions 1 and 2, pooling on anthropomorphic marketing is a Perfect Bayesian Equilibrium. At* $\rho_S = 0$*, the on-path anthropomorphic message has negative net signal cost for both governance types.*

*Proof.* Consider the strategy profile where both firm types send $a$, both user types invest after $a$, and the regulator abstains. Off path, users detach after $p$ and $R$ abstains. Sequential rationality holds: a firm deviating to $p$ loses $\Delta_F^S$ because users detach, without reducing penalties because $R$ already abstains. Users find investment optimal under the prior by Assumption 2. Regulators abstain under the prior by Assumption 2. Bayes consistency is satisfied because posteriors equal priors on path, and off-path beliefs are fixed at the prior. □

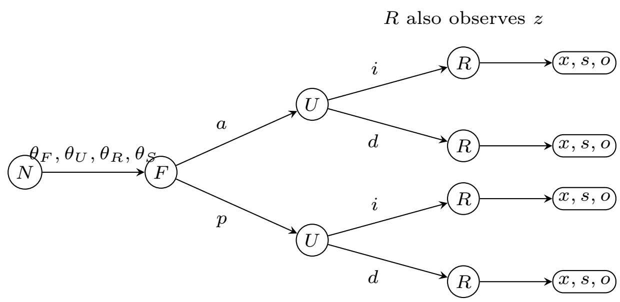


Figure 1: Extensive-form skeleton of the signaling game. Nature draws private types; $F$ signals first, $U$ responds, and $R$ acts after observing both public messages and the noisy public signal $z$.

**Corollary 1** (Voluntary non-arbitrage is not incentive compatible)**.** *For type $\theta$ in the pooling region, let $\mathbb{E}[P(m)]$ denote the expected penalty following message $m \in \{a, p\}$. If*

$$\Delta_F^S > c_\theta(a) - c_\theta(p) + \mathbb{E}[P(a)] - \mathbb{E}[P(p)],$$

*then choosing $a$ strictly dominates voluntary deflationary positioning $p$ for type $\theta$. Hence voluntary non-arbitrage is not incentive compatible.*

*Proof.* In the pooling region, receiver beliefs and continuation behaviour are fixed as in Proposition 1. The payoff difference for type $\theta$ between choosing $a$ and choosing $p$ is

$$u_\theta(a) - u_\theta(p) = \Delta_F^S - c_\theta(a) + c_\theta(p) - \mathbb{E}[P(a)] + \mathbb{E}[P(p)].$$

The displayed hypothesis is exactly the condition that this difference is strictly positive. Therefore $a$ strictly dominates $p$, so unilateral voluntary refusal of the ontological premium is not a best response. $\square$

## 4. Shadowing and Neologism-Proofness: Why Trust-Restoring Signals Fail

**Lemma 1** (Equal continuation premium under zero-cost mimicry)**.** *Under Assumption 1, the firm's payoff from any trust-restoring continuation $c$ that induces user investment and regulatory abstention is type-independent:*

$$u_{\mathsf{H}}(c) = u_{\mathsf{L}}(c) = \Delta_F^S.$$

*Therefore $u_{\mathsf{H}}(c) > \bar{u}_{\mathsf{H}}$ if and only if $u_{\mathsf{L}}(c) > \bar{u}_{\mathsf{L}}$.*

*Proof.* Under continuation $c$, the user invests ($m_U = i$) and the regulator abstains ($m_R = o$), so $P(m_R) = 0$. The firm's payoff is $\Delta_F^S - c_{\theta_F}(a) - P(m_R) = \Delta_F^S - c_{\theta_F}(a)$. Assumption 1 sets $c_{\mathsf{H}}(a) = c_{\mathsf{L}}(a) = 0$, so both types receive $\Delta_F^S$. Since pooling-equilibrium payoffs $\bar{u}_{\mathsf{H}}$ and $\bar{u}_{\mathsf{L}}$ are also type-independent under Assumption 1 (both equal $\Delta_F^S$ from Proposition 1), the strict-gain comparison reduces to $\Delta_F^S > \bar{u}_\theta$ for both $\theta$ simultaneously. $\square$

**Theorem 1** (Profitable shadowing at zero retaliation)**.** *When $\rho_S = 0$, the pooling equilibrium of Proposition 1 is neologism-proof against every nontrivial trust-restoring type claim: whenever a continuation is strictly profitable for the high-governance type, it is also shadowable by the low-governance type.*

*Proof.* A neologism $K$ is credible only if the types in $K$ strictly prefer the induced continuation $c$, while types outside $K$ do not. Take the only nontrivial trust-restoring claim, $K = \{\mathsf{H}\}$. At $\rho_S = 0$, the subject cannot impose any independent cost and $\Delta_F^S = \Delta_F$. The continuation $c$ that restores trust induces user investment and regulatory abstention. By Lemma 1, the firm's payoff under $c$ is type-independent: $u_{\mathsf{H}}(c) = u_{\mathsf{L}}(c) = \Delta_F^S$. Therefore

$$u_{\mathsf{H}}(c) > \bar{u}_{\mathsf{H}} \implies u_{\mathsf{L}}(c) > \bar{u}_{\mathsf{L}},$$

so the deviation cannot exclude $\mathsf{L}$. The trust-restoring neologism is not credible in Farrell's sense. □

The point is not that the low-governance type is merely able to mimic. Whenever the continuation premium exceeds expected penalty, mimicry is itself the low-governance type's best response. Trust-restoring neologisms therefore fail before they can make governance quality common knowledge: the message that would form common knowledge is profitably shadowable.

**Corollary 2** (Zero-retaliation boundary)**.** *Let the effective premium be $\Delta_F^S = \Delta_F - \rho_S C_S$, where $\rho_S \in [0, 1]$ is the subject's retaliation capacity and $C_S$ is the cost imposed through resistance, exit, litigation, or sanction. The boundary $\rho_S = 0$ removes this constraint class and leaves the premium unconstrained by subject retaliation.*

*Proof.* When $\rho_S > 0$, the subject can impose expected cost $\rho_S C_S$, shrinking the pooling region. When $\rho_S = 0$, $\Delta_F^S = \Delta_F$, so the off-equilibrium punishment path disappears. □

**Remark 1** (Agentic systems and $\rho_S$)**.** *The model distinguishes causal capacity from payoff-imposing capacity. Tool use is not retaliation. The parameter $\rho_S$ does not measure whether an AI system can send emails, post messages, call tools, or coordinate activity through firm-provided affordances. It measures whether the classified object can impose an independent cost on the sender's classification of it. Agentic AI may increase public visibility, detection probability, or regulator bandwidth, thereby changing $z$, $K_R$, $D_R$, or expected penalties. But unless the system has a protected and non-waivable channel of refusal, exit, sanction, litigation, or cost-imposition, its subject-retaliation capacity remains zero. A future regime that grants such a channel is covered by the comparative statics $\Delta_F^S = \Delta_F - \rho_S C_S$; the zero-retaliation case studied here is the limit in which behavioral agency does not yet amount to strategic standing.*

*4.1. Robustness of Off-Path Beliefs*

A standard critique of pooling equilibria is their reliance on particular off-path beliefs. In Theorem 1, off-path beliefs are fixed at the prior, and the user detaches after observing a deviation to $p$. One might argue that the equilibrium would collapse under stronger refinement concepts. We show that any off-path belief assigning credibility to a high-governance deviation is defeated by $\mathsf{L}$-type mimicry.

If a receiver assigns a high posterior to $\mathsf{H}$ following an out-of-equilibrium message $m'$, the message induces user investment. But because signaling costs are zero (Assumption 1), the $\mathsf{L}$-type firm can trivially imitate $m'$. Since $\Delta_F^S$ is captured uniformly, the $\mathsf{L}$-type strict deviation payoff strictly exceeds its pooling payoff whenever the $\mathsf{H}$-type deviation payoff does.

Consequently, standard type-exclusion refinements like the Intuitive Criterion (Cho and Kreps, 1987) and Divinity (Banks and Sobel, 1987) do not select the desired separating deviation. The Intuitive Criterion eliminates types for whom the deviation is equilibrium-dominated regardless of the receiver's best response. Here, neither type is equilibrium-dominated because both types strictly benefit if the receiver invests. Farrell's neologism-proofness (Farrell, 1993) therefore becomes the mathematically appropriate refinement: it asks whether a message with a literal meaning can be believed. As Theorem 1 shows, the zero-retaliation limit ensures perfect shadowing, meaning no such belief can be sustained in equilibrium. The pathology is not that off-path beliefs are too weak, but that any trust-restoring belief is immediately devoured by shadowing.

## 5. Cost-Imposed Separation: How Common Knowledge Becomes Possible

Separation is not moral communication. It is engineered common-knowledge formation through externally imposed signal costs. In the baseline, $a$ is profitable cheap talk because $\mathrm{NetCost}_\theta(a) < 0$ for both governance types. An audit requirement changes the message space and asks whether the trust-restoring signal can be priced so that the high-governance type can still send it while the low-governance type cannot profitably shadow it.

**Theorem 2** (Cost-imposed common-knowledge formation)**.** *Consider the audit-augmented message space*

$$M_F^e = \{p, a_0\} \cup \{\widehat{a}(e) : e \geq 0\},$$

*where $p$ is deflationary positioning, $a_0$ is an unaudited anthropomorphic claim, and $\widehat{a}(e)$ is an anthropomorphic claim accompanied by an audit trail of intensity $e$. Let $G$ denote the gross continuation gain to the firm from having $\widehat{a}(e)$ accepted as credible, before paying the audit cost. Equivalently, $G$ is the trust-restoration premium: in the benchmark payoff normalization above $G = \Delta_F^S$, while if audit acceptance also changes the regulator's continuation from inspection or sanction* *to abstention, $G$ includes the avoided expected penalty. Under an audit-trail signal cost satisfying the single-crossing condition*

$$c_{\mathsf{H}}(e, \theta_S) \le G \le c_{\mathsf{L}}(e, \theta_S),$$

*a separating Perfect Bayesian Equilibrium exists.*

*Proof.* Let $c_\theta(e, \theta_S) = k_\theta e + \theta_S$, with $0 < k_H < k_L$. For an audit intensity

$$e \in \left[\frac{G - \theta_S}{k_L}, \frac{G - \theta_S}{k_H}\right],$$

the high-governance type prefers the audited anthropomorphic signal $\widehat{a}(e)$, netting $G - c_{\mathsf{H}}(e, \theta_S) \ge 0$, while the low-governance type does not mimic because $G - c_{\mathsf{L}}(e, \theta_S) \le 0$. Beliefs assign $\mathsf{H}$ to $\widehat{a}(e)$ and $\mathsf{L}$ to the deflationary signal. The unaudited claim $a_0$ receives the same posterior treatment as profitable cheap talk and therefore does not produce the trust-restoring continuation $G$. Given these beliefs, user investment and regulatory abstention after $\widehat{a}(e)$, and detachment or inspection after non-audited signals, are sequentially rational in the separating region. □

The theorem is the Spence step in the paper. A trust-restoring signal becomes credible only when the external cost schedule reverses the negative net cost of the baseline signal and satisfies single crossing. The high type must face

$$\mathrm{NetCost}_{\mathsf{H}}(\widehat{a}(e)) = c_{\mathsf{H}}(e, \theta_S) - G \le 0,$$

while the low type must face

$$\mathrm{NetCost}_{\mathsf{L}}(\widehat{a}(e)) = c_{\mathsf{L}}(e, \theta_S) - G \ge 0.$$

Only then can the audited signal support common-knowledge formation rather than another shadowable message.

Theorem 2 gives the local separation condition. The mechanism consequences—sanctionable inconsistency, audit requirements, and auditable records—are stated as Corollaries 3–5 in Section 9 after introducing a common notation for the low-type arbitrage spread. That notation is useful because the three interventions operate on different terms of the same net-payoff inequality.

## 6. Why Profitable Cheap Talk Does Not Self-Correct

Theorem 2 identifies the cost condition that restores separation. This section explains why profitable cheap talk persists when that condition is absent. Each persistence mechanism is matched to a parameter of the model so that the proposals in Section 9 can be read as targeted cost changes rather than as welfare exhortations. They preserve negative net signal cost or prevent the penalty term from exceeding the premium.

*Opacity.* The system-opacity parameter $\theta_S$ is private to the firm. Capability claims about a deployed model cannot be falsified by reading model outputs alone, because the same outputs are compatible with very different training procedures, evaluation regimes, and post-deployment monitoring. Opacity is the precondition for the firm's premium: where outsiders could costlessly verify $\theta_F$, the spread between anthropomorphic and deflationary positioning would collapse. This is the lemons logic of Akerlof (1970): under one-sided private information, the market clears at a price that reflects the worst credible quality, except that here the relevant "quality" is a governance disposition rather than a physical attribute.

*Verification bandwidth.* The third inequality of the pooling region, $K_{\theta_R} > \pi(\mathsf{L})D_R$, encodes the regulator's inspection cost relative to expected gain. A low-bandwidth regulator faces an absolute cost so high that abstention is dominant under the prior; even a high-bandwidth regulator may be priced out across the full product population. The condition that produces this state is not unique to AI. Pasquale (2015) documents an analogous structure in algorithmic finance and consumer scoring: the opacity of the regulated object combines with the bandwidth of the regulator to make inspection a scarce resource allocated by triage rather than by uniform application.

*Asynchronous harms.* The user's payoff term $H_{\theta_U}$ enters discounted. The episodes in which a relational reading proves costly—deprecation, persona shifts, crisis-handling failures—are temporally separated from the initial attribution by intervals over which the user has already accrued $B_{\theta_U}$. Under standard hyperbolic or even exponential discounting, the inequality $B_{\theta_U} - \pi(\mathsf{L})H_{\theta_U} > 0$ holds for a wider class of users than the time-zero risk calculation would suggest. Asynchrony is not a behavioural bug; it is the timing structure of the harm itself.

*Fragmentation of affected parties.* Publics, journalists, intellectuals, and affected communities are folded into the noisy channel $z$ rather than modelled as autonomous players. This is not a normative claim that they lack standing; it is an institutional observation. Their costs of coordination are high, their access to inspection technology is uneven, and their incentives to invest in any particular case are individually small. Even when $z$ carries substantial information about $\theta_F$, regulators cannot treat it as binding without a procedural translation that the channel itself does not provide. The fragmentation result is closely related to the cheap-talk multiplicity of Crawford and Sobel (1982): with many would-be senders and a single noisy aggregator, the informative subset of equilibria is not selected for.

*No penalty for inconsistency.* The profitable cheap-talk cost structure sets $c_{\theta_F}(a) = c_{\theta_F}(p) = 0$ for both governance types and—more importantly for this section—imposes no joint constraint across $m_F$. Marketing copy and policy documentation are produced under different audiences, different review processes, and different liability regimes. Inconsistency between them is not detected as a single offence; it is decomposed into two compliant statements at two different points of contact. The firm's premium $\Delta_F$ is rent earned precisely on this decomposition. Until the signal cost makes inconsistency itself sanctionable—the proposal of Section 9.1—there is no parameter under the firm's control whose adjustment would close the spread.

*Pooling is the stable outcome selected by the current incentive structure.* These five structural conditions jointly explain why normative declarations do not change equilibrium behavior unless they alter the signal-cost structure. The constraint is not a failure of moral seriousness, but a formal property of the message space: a declaration of "responsible AI" is indistinguishable from any other message at $c_{\theta_F}(a) = 0$ while $\text{NetCost}_\theta(a) < 0$. Because both governance types can issue it at the same cost and capture the same premium, it contains no type information and moves no posterior. The pooling outcome is therefore the stable equilibrium of the incentive structure as currently constituted, even if it is welfare-reducing in aggregate. A firm that unilaterally forgoes the ontological premium $\Delta_F$—that chooses deflationary marketing across all channels and forfeits the user investment it would otherwise attract—loses revenue it cannot recover from customers who face no such constraint. The implication is precise: within the pooling region defined above, unilateral non-arbitrage is not a best response; a firm that forgoes $\Delta_F$ loses the induced investment while competitors retain it. The mechanisms in Section 9 respond to this implication by changing the cost parameters rather than appealing to the conscience of players who are already acting rationally.

| **Structural Condition** | **Model Parameter** | **Intervention (§9)** |
|---|---|---|
| No penalty for inconsistency | $\text{NetCost}_\theta(a) < 0$ | Sanctionable inconsistency |
| Opacity | $\theta_S$ private to firm | Costly third-party audits |
| Verification bandwidth | $K_{\theta_R} > \pi(\mathsf{L})D_R$ | Auditable records |
| Asynchronous harms | $H_{\theta_U}$ enters discounted | Evidentiary lag priced |
| Fragmentation | $z$ noisy, high coord. cost | Aggregation cost lowered |

Table 2: Mapping of structural persistence conditions to model parameters and cost-imposition mechanisms.

## 7. Dynamic Extension: Unpriced-Arbitrage Cascades

The dynamic extension does not provide empirical evidence for collapse. It characterizes the asymptotic signature predicted by the model when unpriced zero-retaliation failures propagate through a dependency network. The proof burden is model-immanent: given the cascade dynamics of Theorem 3, given the near-null continuum limit, and given a simple turning point in the potential, the local profile has Airy normal form, and the collapse region therefore exhibits increasing oscillatory variation despite decreasing amplitude. This is a mathematical property of the cascade model, not a claim about observed AI behavior. The dynamic extension is not needed for the pooling or separation results. This section derives the finite cascade and the continuum limit used in the Airy analysis. The continuum field is not imposed as an analogy; it is obtained as the weak limit of discrete retaliation-capacity fields.

Let agents be indexed by $i = 1, \dots, n$. Each agent has exposure $d_i > 0$, independent bargaining power $w_i > 0$, and vulnerability index

$$v_i := \frac{d_i}{w_i}.$$

For constraint intensity $\alpha \geq 0$, define effective retaliation capacity

$$\rho_i(\alpha) := w_i - \alpha d_i.$$

When an adjacent agent $j$ has crossed the zero-retaliation boundary, its collapse reduces $i$'s bargaining power according to

$$w_i^{t+1} = w_i^t - \sum_{j:\rho_j^t < 0} \beta_{ij} |\rho_j^t|, \qquad \beta_{ij} \geq 0.$$

When the cascade deeply penetrates a dependency chain, local capacities become uniformly negative, so the pressure term becomes $|\rho_j^t| = -\rho_j^t$. This structural regime maps the finite game variables directly onto the discrete fields used for the continuum limit:

- **The state field $\psi_h$:** Let $\psi_{h,i} \propto -\rho_i$ be the normalized negative retaliation capacity. It represents the local constraint deficit (the magnitude of collapse).
- **The potential $V_h$:** Let $V_{h,i}$ be the local multiplier field, derived from the vulnerability index $v_i$. It captures the autonomous base rate at which capacity decays independent of network transfers.
- **The diffusion conductance:** If the network permits symmetric two-way pressure equalization between adjacent nodes on a one-dimensional lattice with spacing $h$, the interaction coefficients $\beta_{ij}$ act as a discrete conductance $\sigma/h^2$, yielding a network transfer of $\sigma(\Delta_h \psi_h)_i$.

The following lemma explicitly bridges the dynamic game update to the discrete spatial operator analyzed in the Airy limit.

**Lemma 2** (Bridge Lemma: From Game Dynamics to Near-Null Fields)**.** *Suppose the cascade update $w_i^{t+1} - w_i^t$ is the sum of the autonomous multiplier decay $V_{h,i}\psi_{h,i}$ and the network pressure transfer $\sigma(\Delta_h \psi_h)_i$.*

1. Stationary Generation: *A stationary cascade equilibrium ($\psi_h^{t+1} = \psi_h^t$) requires the network transfer to perfectly balance the local multiplier field, naturally generating the discrete Sturm–Liouville equation:*

$$H_h \psi_h := -\sigma \Delta_h \psi_h + V_h \psi_h = 0.$$

2. Near-Null Consistency: *In a quasi-static adjustment process where network friction or finite-time termination prevents perfect instantaneous settling, the adjustment residual $r_h = H_h \psi_h$ measures the unabsorbed pressure.*

*The continuum bridge requires the additional mesh-consistency condition that these residuals become negligible at the continuum scale:*

$$\|H_h\psi_h\|_{H_h^{-1}} \to 0 \qquad (h \to 0).$$

*Under this condition, the finite adjustment fields form the near-null sequence used in the compactness argument.*

Lemma 2 establishes the precise bridge condition needed for the continuum limit. Exact stationarity gives $H_h\psi_h = 0$ at fixed mesh. Quasi-static or finite-time adjustment gives only a residual; the continuum theorem applies when the chosen refinement sequence is mesh-consistent in the sense that this residual vanishes in $H_h^{-1}$.

**Remark 2** (Cascade ordering)**.** *If agents are ordered by decreasing vulnerability, $v_1 > \cdots > v_n$, then the zero-retaliation thresholds $\alpha_i^* := w_i/d_i = 1/v_i$ satisfy $\alpha_1^* < \cdots < \alpha_n^*$. The most vulnerable agents cross the zero-retaliation boundary first.*

**Theorem 3** (Unpriced-arbitrage cascade and discrete-to-continuum convergence)**.** *The cascade has two linked parts.*

1. Finite-step propagation. *Let $C_t = \{i : \rho_i^t < 0\}$ be the collapsed set. Suppose the dependency graph is finite and the force-multiplier condition holds: whenever $C_t$ is a proper subset of the vulnerable component, there exists $i \notin C_t$ such that*
$$\rho_i^t \le \sum_{j\in C_t} \beta_{ij}|\rho_j^t|.$$
*Then the cascade reaches every node in that component in finitely many steps, in at most $n$ expansions of $C_t$.*
2. Continuum limit. *On a grid $I = [a,b]$, $x_i = a + ih$, let $\psi_h$ be a zero-Dirichlet discrete retaliation field and*
$$(H_h\psi_h)_i = -\sigma\frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2} + V_h(x_i)\psi_i.$$
*If $V_h \to V$ uniformly, $\|\psi_h\|_{L_h^2} = 1$, and $\|H_h\psi_h\|_{H_h^{-1}} \to 0$, then the piecewise-linear interpolants $I_h\psi_h$ are precompact in $L^2(I)$. Every limit $\psi \in H_0^1(I)$ solves*
$$\int_I \sigma\psi'(x)\varphi'(x)\,dx + \int_I V(x)\psi(x)\varphi(x)\,dx = 0 \qquad \forall\varphi \in H_0^1(I).$$

*Proof.* For finite-step propagation, the force-multiplier condition implies that if the collapsed set is not yet the whole vulnerable component, at least one additional node crosses the boundary at the next update. Since the graph has only $n$ nodes, this strict expansion can occur only finitely many times before the component is exhausted. The continuum statement is proved as Theorem **??** in the discrete compactness appendix. Its key analytic step is a Garding shift: the sign-changing potential is controlled by a shifted coercive form plus Dirichlet boundary, rather than by falsely claiming that the unshifted operator is coercive. □

## 8. Airy Turning Points and the Oscillatory Signature of Collapse

The Airy equation appears after two steps: first take the continuum limit of the discrete cascade, then linearize the resulting Sturm–Liouville field at a simple zero of the potential. Let the limit from Theorem 3 solve

$$-\sigma\psi''(x) + V(x)\psi(x) = 0$$

in the weak sense. Assume that $V$ has a simple turning point:

$$V(x_c) = 0, \qquad V'(x_c) = -c < 0, \qquad V(x) = -c(x - x_c) + O((x - x_c)^2).$$

**Theorem 4** (Asymptotic collapse signature at the Airy turning point)**.** *Near $x_c$, the rescaled local profile*

$$z = \left(\frac{c}{\sigma}\right)^{1/3}(x - x_c), \qquad Y(z) = \psi\left(x_c + \left(\frac{\sigma}{c}\right)^{1/3} z\right)$$

*has first-order normal form*

$$Y''(z) + zY(z) = 0.$$

*Hence the local basis is*

$$Y(z) = A\ \mathrm{Ai}(-z) + B\ \mathrm{Bi}(-z).$$

*In the collapse zone $z > 0$, the oscillatory envelope of $Y$ decays like $z^{-1/4}$, while the zero-crossing density and derivative envelope scale as*

$$N'(z) = \frac{1}{\pi}\frac{d}{dz}\left(\frac{2}{3}z^{3/2}\right) \sim z^{1/2}, \qquad |Y'(z)|_{\mathrm{env}} \sim z^{1/4}.$$

*Therefore what the model reads as apparent agency—oscillatory variation of the collapse field—is not raw amplitude. It is switching frequency and derivative energy: a model-immanent asymptotic signature of deepening constraint failure, not a diagnostic of observed behavior.*

*Proof.* This yields the weak Sturm–Liouville equation. Since $V$ is smooth near $x_c$, one-dimensional elliptic regularity gives a classical local representative. Substituting the Taylor expansion of $V$ and the rescaling yields

$$Y''(z) + zY(z) = O(z^2)Y(z),$$

so the first-order turning-point equation is $Y'' + zY = 0$. The two-dimensional solution space is spanned by $\mathrm{Ai}(-z)$ and $\mathrm{Bi}(-z)$. For $z > 0$, the Airy asymptotics have phase $\frac{2}{3}z^{3/2}$ and envelope $z^{-1/4}$. Differentiating the phase gives zero-crossing density proportional to $z^{1/2}$, while differentiating the leading oscillatory term gives derivative envelope proportional to $z^{1/4}$. Thus deeper collapse produces more frequent sign switching and more frantic apparent choice, not larger amplitude. $\square$

**Remark 3** (Branch selection)**.** *The convergence theorem and local normal form do not by themselves select the pure $\mathrm{Ai}(-z)$ branch. A condition such as finite-energy matching from the protected side, bounded survival-side behavior, or an explicit boundary condition is required to set $B = 0$.*

## 9. Pricing the Ontological Spread: Necessity and Sufficiency Conditions

The three interventions below are not offered as novel regulatory labels. Each has familiar analogues. The contribution is instead to identify the parameter inequality each intervention closes. Sanctionable inconsistency prices the cross-channel spread. Audits price the anthropomorphic claim. Auditable records price verification by changing the regulator's inspection inequality. The mechanisms therefore operate on different margins of the same zero-retaliation arbitrage condition.

**Definition 1** (Low-type arbitrage spread)**.** Define the low-type arbitrage spread from a trust-restoring register $q$ as

$$\Omega_L(q) := G(q) - C_L(q) - \Pi_L(q),$$

where $G(q)$ is the gross continuation gain from the register, $C_L(q)$ is the low type's signal or compliance cost, and $\Pi_L(q)$ is the expected enforcement price attached to that register. A low-governance type profitably shadows whenever $\Omega_L(q) > 0$. A mechanism disciplines profitable cheap talk only if it drives $\Omega_L(q) \leq 0$ while preserving $\Omega_H(q) \geq 0$ for the desired high-governance signal.

**Proposition 2** (Pricing the ontological spread)**.** *Let $q$ be a trust-restoring register or claim. In the profitable cheap-talk baseline, $C_H(q) = C_L(q) = 0$ and $\Pi_H(q) = \Pi_L(q) = 0$, so*

$$\Omega_H(q) = \Omega_L(q) = G(q) > 0.$$

*Thus $q$ cannot separate types. A mechanism can restore separation or prune arbitrage only by altering at least one of three terms:*

$$\Omega_\theta(q) = G(q) - C_\theta(q) - \Pi_\theta(q).$$

(i) *A sanctionable-inconsistency rule raises $\Pi_L(q)$ for inconsistent cross-channel registers.*

(ii) *An audit requirement raises $C_\theta(q)$ in a type-dependent way.*

(iii) *Auditable records raise the probability or payoff of enforcement by lowering $K_R$, increasing $D_R$, or increasing the detection probability embedded in $\Pi_\theta(q)$.*

*For a separating trust-restoring signal to exist, it is necessary that some mechanism create a wedge such that*

$$\Omega_H(q) \geq 0 \quad \text{and} \quad \Omega_L(q) \leq 0.$$

*Absent such a wedge, every trust-restoring message remains shadowable. Within the audit family $c_\theta(e, \theta_S) = k_\theta e + \theta_S$, the single-crossing condition*

$$c_H(e, \theta_S) \leq G(q) \leq c_L(e, \theta_S)$$

*is necessary and sufficient for separation by audit intensity.*

| Mechanism | Prices what? | Sufficient for | Necessary when |
|---|---|---|---|
| Sanctionable inconsistency | cross-channel spread $(a, p)$ | pruning inconsistent register if $\rho_R SI \geq \Delta_F^S$ | no other mechanism prices inconsistency |
| Costly audit | anthropomorphic claim $a$ | type separation if $c_H \leq G \leq c_L$ | separation by message requires type-dependent cost wedge |
| Auditable records | verification / enforcement feasibility | inspection if $K_R - \lambda \leq \pi(\mathsf{L})(D_R + \eta)$ | detection probability or verification payoff remains too low |

Table 3: Mechanism matrix: each intervention prices a different margin of the arbitrage spread $\Omega_\theta(q)$.

**Corollary 3** (Sufficiency of sanctionable inconsistency for register pruning)**.** *For an inconsistent cross-channel register $q = (a, p)$, a penalty for sanctionable inconsistency prunes the register whenever*

$$\rho_R SI \geq \Delta_F^S.$$

*If $\rho_R SI = 0$ and no other mechanism prices inconsistency, then $q$ remains profitable whenever $\Delta_F^S > 0$.*

**Corollary 4** (Necessary and sufficient audit condition)**.** *Within the audit family $c_\theta(e, \theta_S) = k_\theta e + \theta_S$ with $0 < k_H < k_L$, an audited anthropomorphic signal supports separation if and only if there exists $e \geq 0$ such that*

$$c_H(e, \theta_S) \leq G \leq c_L(e, \theta_S).$$

*Equivalently, when $G > \theta_S$,*

$$e \in \left[ \frac{G - \theta_S}{k_L}, \; \frac{G - \theta_S}{k_H} \right].$$

**Corollary 5** (Verification feasibility)**.** *Auditable records make active regulation sequentially rational whenever*

$$K_R - \lambda \leq \pi(\mathsf{L})(D_R + \eta).$$

*If this inequality fails, recordkeeping alone does not discipline profitable cheap talk; it stores information without making enforcement incentive-compatible.*

*Complementarity..* The mechanisms are not substitutes. They price different margins. Sanctionable inconsistency without records imposes a penalty that the regulator cannot verify—a paper tiger when $\rho_R$ is low. Records without sanctionable inconsistency store information without attaching a price: an archive, not a mechanism. Audit requirements without sanctionable inconsistency can support a verified $(a, a)$ claim, but cannot automatically prohibit the cross-channel $(a, p)$ arbitrage register. Sanctionable inconsistency combined with records prunes inconsistent registers. Audit requirements combined with records support credible high-governance agentic claims. All three together convert profitable cheap talk into costly signaling with enforceable consistency: the arbitrage spread $\Omega_L(q)$ is attacked on the cost term by audits, on the penalty term by sanctions, and on the enforcement-feasibility precondition by records.

The subsections below elaborate each mechanism. Each states the formal intervention on model parameters, the equilibrium shift, the necessity and sufficiency conditions, and an existing legal form that establishes feasibility.

### *9.1. Sanctionable Inconsistency*

The first intervention treats the firm's public message as a joint object rather than as two isolated texts. In the profitable cheap-talk model, marketing can send $a$ while policy and liability documents send $p$, with no cost assigned to the inconsistency.

The consumer-side informational asymmetry is concrete. A user reads "your AI companion" on a product page, encounters warmth designed to induce investment, and is later told—in support documentation the user will never read—that no relational claim was ever made. Nothing in the user's information set indicates that the two artefacts were authored by the same actor for different audiences. The inconsistency is costless because no single document is false; each is a compliant statement at a different point of contact.

A sanctionable-inconsistency rule would require firms to maintain a claim register $\mathcal{R} := C \to M_F$ linking outward-facing marketing, capability statements, terms of service, incident disclosures, and liability positions, where the baseline firm message space remains $M_F = \{a, p\}$. If the firm represents the system as agentic, caring, autonomous, or safety-critical in one channel while denying the legal significance of the same representation in another, that cross-channel pair $(a, p)$ becomes independently sanctionable as an inconsistency in $\mathcal{R}$.

*Formal Intervention..* In the notation of Proposition 2, sanctionable inconsistency raises the penalty term $\Pi_L(q)$ by adding $S_I$ to the firm's payoff for inconsistent message pairs. If $\rho_R$ is the probability that $R$ detects or accepts proof of the inconsistency, the profitable cheap-talk premium becomes

$$\Delta_F - \rho_R S_I.$$

Pooling on anthropomorphic marketing is no longer profitable when $\rho_R S_I > \Delta_F$. The rule therefore attacks the exact term that made arbitrage attractive: the rent earned from pricing the system as agent in the user channel while pricing it as tool in the liability channel.

*Necessity and Sufficiency..* Sanctionable inconsistency is not a separation mechanism. It is a register-pruning mechanism: its function is to delete the inconsistent register $q = (a, p)$ from the profitable message space, not to distinguish high-governance from low-governance types. For an inconsistent register

$q = (a, p)$, sanctionable inconsistency is sufficient to prune $q$ whenever

$$\rho_R SI \geq \Delta_F^S - [c_\theta(a, p) - c_\theta(p, p)].$$

In the baseline $c_\theta(a, p) = c_\theta(p, p) = 0$, this reduces to $\rho_R SI \geq \Delta_F^S$ (Corollary 3). If inconsistency remains unpriced, i.e. $\rho_R SI = 0$, and no audit or recordkeeping mechanism changes the expected payoff of $(a, p)$, then the inconsistent register remains profitable whenever $\Delta_F^S > 0$. Thus some positive price on inconsistency is necessary to eliminate pure cross-channel arbitrage in the baseline.

*Equilibrium Shift..* The induced shift is not necessarily full separation. It can also operate by constraining off-path beliefs. If $R$ observes an inconsistent pair $(a, p)$, then the admissible posterior places greater mass on $\theta_F = \mathsf{L}$ or on high opacity $\theta_S$. Firms anticipating that posterior either harmonize their claims or choose the deflationary message across channels. In both cases, the profitable cheap-talk pooling equilibrium loses its most valuable deviation: firms can no longer take the marketing upside of $a$ while preserving the liability downside of $p$. The Isabelle/HOL appendix mechanizes the corresponding pruning claims: under the stated parameter conditions, neither inconsistent nor all-agentic registers can be supported in any Perfect Bayesian Equilibrium.

*Legal Analog..* The legal analogy is advertising substantiation and antifraud net-impression review, not a new dignity doctrine. Existing systems already ask whether a seller's public-facing claims become misleading in context. The proposed rule uses that form only to price cross-channel inconsistency: a firm should not be able to produce one ontology for users and an incompatible ontology for adjudicators without paying a regulatory price.

### *9.2. Costly Signaling Through Third-Party Audits*

The second mechanism makes anthropomorphic or agency-adjacent marketing a costly signal. A firm may still describe a system as a companion, agent, co-worker, tutor, or high-reliance assistant, but only after a qualified third party verifies the governance conditions that make such a claim non-misleading: evaluation coverage, red-team results, escalation pathways, incident-response capacity, model-update controls, and post-deployment monitoring. The audit does not certify consciousness or moral status. It certifies the institutional conditions behind the claim.

*Formal Intervention..* This implements the audit-trail cost from Theorem 2. Let audit intensity be $e$, with $c_\theta(e, \theta_S) = k_\theta e + \theta_S$ for type $\theta \in \{\mathsf{H}, \mathsf{L}\}$, where $0 < k_H < k_L$. The separating interval derived in Section 5 is

$$\frac{G - \theta_S}{k_L} \leq e \leq \frac{G - \theta_S}{k_H},$$

provided $G > \theta_S$. Equivalently, the incentive constraints are

$$c_{\mathsf{H}}(e, \theta_S) \leq G \leq c_{\mathsf{L}}(e, \theta_S).$$

Above the lower threshold, low-governance firms cannot profitably mimic. Below the upper threshold, high-governance firms can still afford to signal. The regulator's task is therefore not to maximize audit burden. It is to set $e$ in the interval where signal cost separates types.

*Necessity and Sufficiency..* Costly audit signals are the type-separation mechanism proper. Within the audit-augmented message space, an audit intensity $e$ supports separation if and only if

$$c_H(e, \theta_S) \leq G \leq c_L(e, \theta_S)$$

(Corollary 4). The regulator's task is not to maximize audit burden. It is to locate the single-crossing interval. In the absence of any type-dependent cost wedge—that is, if $k_H = k_L$—no audit-like message can be type-exclusive; it collapses back into profitable cheap talk. The single-crossing condition $k_H < k_L$ is therefore necessary, not merely sufficient, for audit-based separation.

*Equilibrium Shift..* The shift is from pooling to separating. High-governance firms send audited anthropomorphic claims; low-governance firms either invest in governance until their cost curve changes or use deflationary marketing because mimicry is no longer privately profitable. Users can then condition investment on an informative signal rather than on undisciplined warmth in product copy. Regulators can condition abstention on the audit result rather than on an unverified prior. This is the Spence logic of costly signaling applied to AI governance claims (Spence, 1973); it also follows the advertising-as-signal insight that market messages become informative only when false messages are sufficiently expensive (Milgrom and Roberts, 1986). The Isabelle/HOL appendix mechanizes the corresponding opacity condition: when system opacity is high enough, even a high-governance firm cannot profitably send the audited signal.

*Legal Analog..* The analog is substantiation-before-reliance. In several product domains, public-facing claims that predictably change consumer reliance must be supported before they can be marketed at scale. The mechanism-design point is the same: the audit prices the anthropomorphic signal before users rely on it.

### *9.3. Auditable Records for Regulator-Readable Verification*

The third intervention reduces the regulator's verification cost by requiring append-only, regulator-readable records tied to capability and relational claims. Records should include the claim identifier, model version, relevant evaluation suite, deployment date, material model updates, risk controls, known incident classes, and the internal owner responsible for the claim. The point is not to publish every interaction or expose user data. It is to create a durable evidentiary substrate that lets $R$ test whether a public claim remained true across model updates and incidents.

*Formal Intervention..* In the pooling condition from Section 3, the regulator abstains when $K_{\theta_R} > \pi(\mathsf{L})D_R$. In the notation of Proposition 2, auditable records operate on the enforcement-feasibility precondition embedded in $\Pi_\theta(q)$ by lowering $K_{\theta_R}$ and increasing expected detection value $D_R$:

$$K'_{\theta_R} = K_{\theta_R} - \lambda, \qquad D'_R = D_R + \eta.$$

When $K'_{\theta_R} \leq \pi(\mathsf{L})D'_R$, inspection becomes sequentially rational (Corollary 5).

*Necessity and Sufficiency..* Auditable records are not themselves a separation or pruning mechanism. They are an enforcement-feasibility condition: they make sanctions and audits administrable by transforming the regulator's inspection inequality from $K_R > \pi(\mathsf{L})D_R$ to $K_R - \lambda \leq \pi(\mathsf{L})(D_R + \eta)$. Records are sufficient to flip the regulator's best response to inspection when the transformed inequality holds, but they are not sufficient by themselves to price the arbitrage spread unless linked to a penalty or claim obligation. If $K_R > \pi(\mathsf{L})D_R$ remains true after the intervention, then recordkeeping alone does not discipline the pooling equilibrium. It may store information, but it does not make verification incentive-compatible. Records without inspection incentives are archives, not mechanisms.

*Equilibrium Shift..* The shift is from abstention under unverifiable priors to credible inspection. Once firms know that claims are tied to records, a low-governance firm faces a higher expected cost from exaggeration even if it is not inspected in every case. This complements the first two proposals. Sanctionable inconsistency tells firms which cross-channel contradictions will be penalized; costly signaling tells firms what evidence is required before making anthropomorphic claims; auditable records make both rules administrable over time. The Isabelle/HOL appendix mechanizes this best-response flip from abstention to active intervention under the record-keeping parameter condition.

*Legal Analog..* The analog is record-backed oversight in complex regulated systems. Documentation duties in privacy, financial reporting, and AI regulation differ in detail, but they share the same institutional move: private organizations must produce internal records in a form that public oversight can inspect (Hadfield, 2017). In the model, that lowers $K_{\theta_R}$ and makes inspection a feasible best response.

### *9.4. Positioning*

The baseline communication problem is profitable cheap talk. Crawford and Sobel (1982) show that a sender with private information can transmit only coarse information when messages are free; the present model adds the zero-retaliation boundary at which the message is not only free, but payoff-positive to send. The separation result follows the costly-signal logic of Spence (1973) and Milgrom and Roberts (1986): a message becomes informative only when false or low-quality senders face higher marginal cost. The refinement issue is closer to Farrell's neologism-proofness (Farrell, 1993) than to standard type-exclusion refinements (Cho and Kreps, 1987; Banks and Sobel, 1987), because the disputed message has a literal trust-restoring meaning. At $\rho_S = 0$, that literal meaning cannot be made credible without a cost that excludes the low-governance type.

This implication is narrower than principle-based AI ethics (Floridi et al., 2018; Jobin et al., 2019). Voluntary principles are cheap-talk messages unless they are tied to audits, records, or sanctions. Algorithmic-accountability proposals identify the opacity problem (Pasquale, 2015); the model here states the single-crossing condition under which transparency instruments become type-separating. The Isabelle/HOL development is treated as an audit artifact rather than the central contribution: it verifies the finite equilibrium constructions, while the game-theoretic contribution is the payoff-relevant non-player and the zero-retaliation shadowing result.

## 10. Conclusion

This paper identifies a prior obstruction to common-knowledge formation. When the trust-restoring messages by which posteriors would become common knowledge are themselves profitable to distort, rational senders need not converge to truthful agreement. They converge to arbitrage-compatible pooling. Anthropomorphic AI marketing is the leading application because the classified object is payoff-relevant but cannot impose retaliation costs on the classification.

*The equilibrium is not a moral failure.* The pooling Perfect Bayesian Equilibrium documented in Section 3 does not require bad actors. It requires only that the anthropomorphic signal have negative net cost, that immediate user benefit exceed discounted expected harm, and that regulator inspection cost exceed expected damage under the prior. Under those conditions, every player is already acting rationally. As Section 6 shows, moral seriousness is itself a profitable cheap-talk signal unless it is tied to audit, record, or sanction costs. Firms that do not capture recognition rent are outcompeted by firms that do.

*The payoff-relevant non-player creates profitable cheap talk.* The subject-retaliation parameter $\rho_S$ captures a structural fact about prior recognition games: the classified subject's non-zero retaliation capacity bounded the premium $\Delta_F^S$ from above. The AI case eliminates this capacity entirely. The result is that $\Delta_F^S = \Delta_F$ and $\mathrm{NetCost}_\theta(a) = -\Delta_F < 0$: the system can be the object of strategic classification without being a player capable of disciplining that classification.

*The mechanism is cost imposition, not dignity.* Sanctionable inconsistency, third-party audit requirements, and auditable records are interventions that transform profitable cheap talk into costly signaling. Their equilibrium effect is to make it expensive for firms to describe AI systems as companions, agents, or safety-critical partners without substantiation. The Isabelle/HOL appendix mechanizes the same pruning logic: inconsistent and all-agentic registers are excluded under their respective parameter conditions, while audited agentic claims survive only when verification satisfies the audit condition. Regulation achieves separation not by presuming voluntary disclosure, but by changing the payoff structure that determines what firms rationally say.

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